



Python Programming Foundations

Week 7 - RBA & Project Framing
Day 1: "Intent Before Creation"

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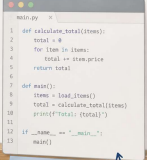
Intent Should Exist Before Generation

- AI can generate code quickly, but speed does not prove the project is clear.
- Before generation, we need to know what the project is for, what it should do, and what decisions should remain human decisions.

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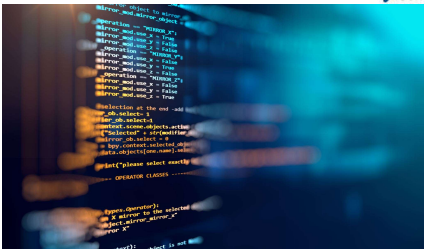
Start with Intent
Define what and why before generating how.

generated code comes later

Intent Before Code

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Review Coding Lab (Opt)

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Review: You Have Several Python Paths Now

You have practiced:

- values and output
- decisions and loops
- functions and collections
- files and structured data
- APIs and architecture recognition

Now the question becomes: *which path fits the project?*

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values/
output

- variables
- expressions
- calculations
- print output
- formatting



decisions/
loops

- conditions
- if / else
- loops
- validation
- control flow



functions/
collections

- define functions
- parameters
- lists / dicts
- iteration
- reuse logic



files/
data

- read / write files
- structured data
- CSV / JSON
- data storage
- data processing



APIs/
architecture

- HTTP requests
- JSON / APIs
- modular design
- separation of concerns
- scalable structure

Which path fits the project?

Review: You Have Several Python Paths Now

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What Counts as Success Today

- define project purpose
- identify inputs and outputs
- name constraints and risks
- choose a likely structure
- set AI-use boundaries
- produce framing that can support capstone approval

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PROJECT FRAMING

— A SUCCESS PATTERN —

- 1 **PURPOSE**
What problem are we solving and why does it matter?
- 2 **INPUTS AND OUTPUTS**
What goes in? What comes out? How will success be measured?
- 3 **CONSTRAINTS AND RISKS**
What limits us? What could go wrong? What assumptions matter?
- 4 **LIKELY STRUCTURE**
What components or modules will this likely need?
- 5 **AI BOUNDARIES**
What should AI help with? What should stay in human control?
- 6 **CAPSTONE-READY FRAMING**
Clear intent, clear plan, clear value. Ready to build and to present.

Good framing now. Strong project later.

Today's Success

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What We Will Use Today

- ▶ purpose
- ▶ inputs
- ▶ outputs
- ▶ constraints
- ▶ possible structure
- ▶ risks
- ▶ AI boundaries

These are project decisions before they are coding decisions.

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TODAY'S FRAMING TOOLS

Use these seven tools to frame any project.



Toolbox

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What We Will "Skip" For Now

- ▶ full build work
- ▶ polishing features
- ▶ publishing packages
- ▶ advanced dependency management
- ▶ asking AI to replace the project plan

Today is framing before building.

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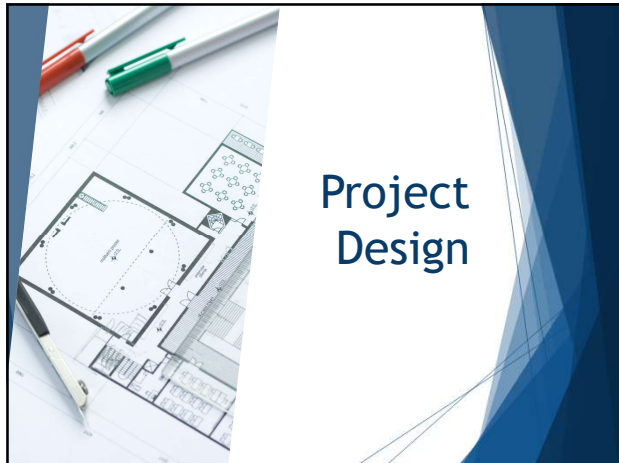
PARKED FOR LATER



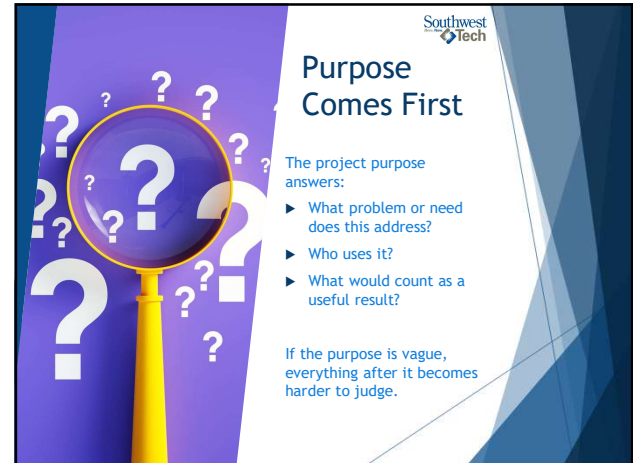
Useful later, not today's target

Parked For Later

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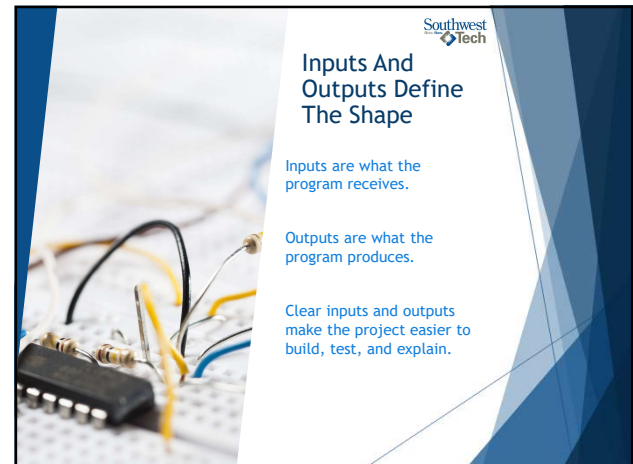
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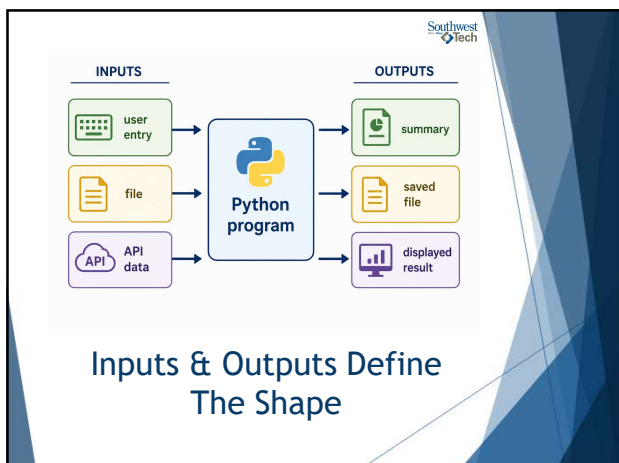
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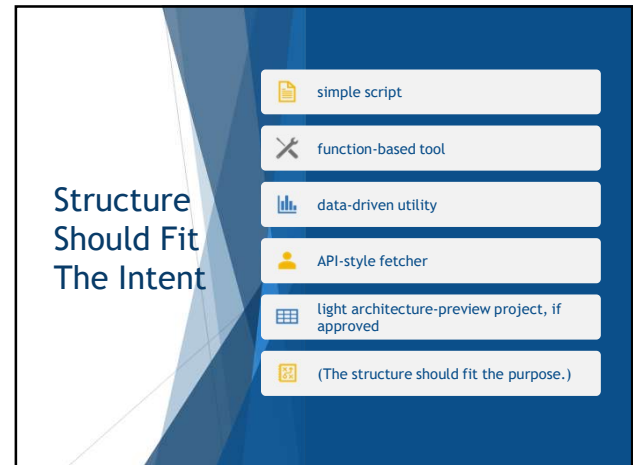


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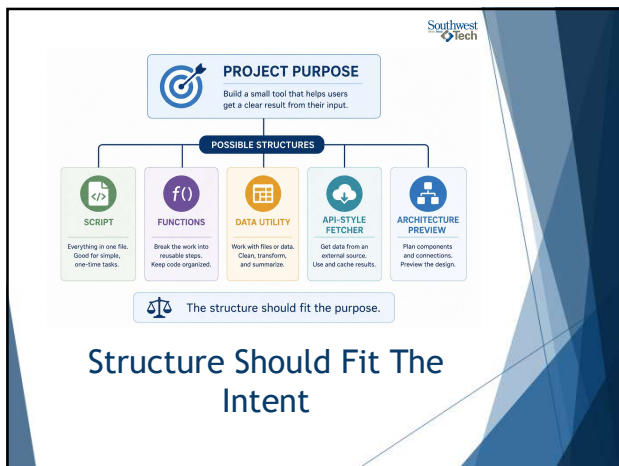


Constraints Are Design Information

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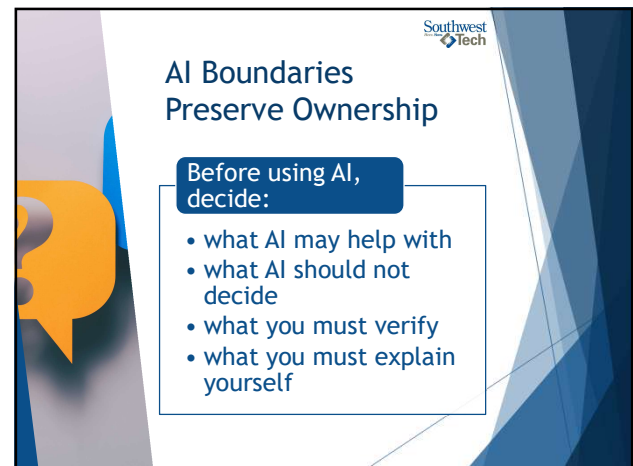


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Structure Should Fit The Intent

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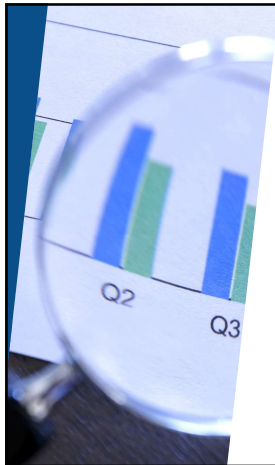


AI Boundaries Preserve Ownership

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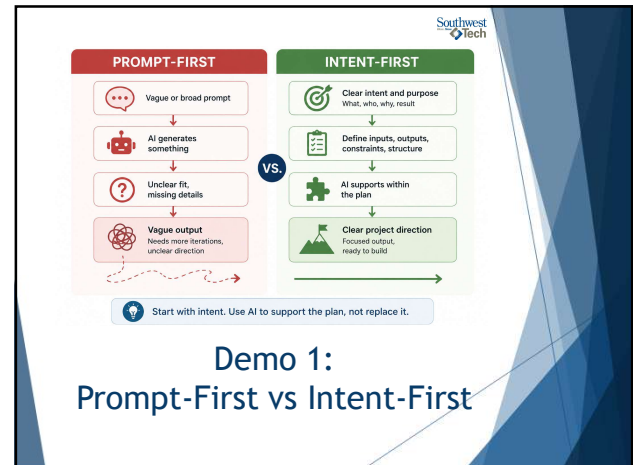


What to Watch For

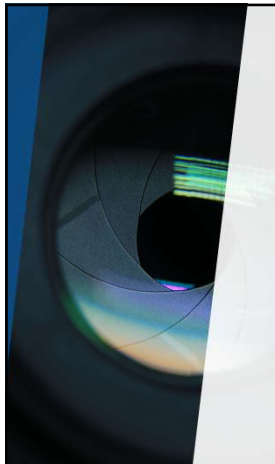
Watch how the project changes when we start with intent instead of a vague prompt.

- ▶ The goal is not a prettier prompt.
- ▶ The goal is better judgment.

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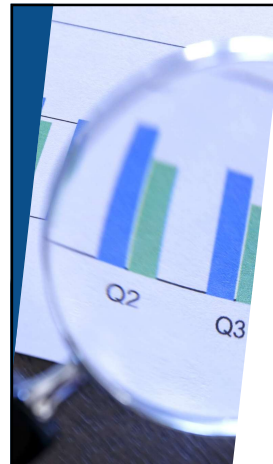
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Demo #2

"Framing A Study Timer"

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What to Watch For

A framing artifact should make the project easier to understand before code exists.

Look for:

- ▶ Purpose
- ▶ Inputs
- ▶ Outputs
- ▶ Constraints
- ▶ Structure
- ▶ AI boundaries.

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Demo 2: Framing A Study Timer

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Common Failure: Vague Ambition

"I want to make something useful"
... is **not** enough.

- ▶ Useful for whom?
- ▶ With what input?
- ▶ Producing what output?
- ▶ Under what limits?

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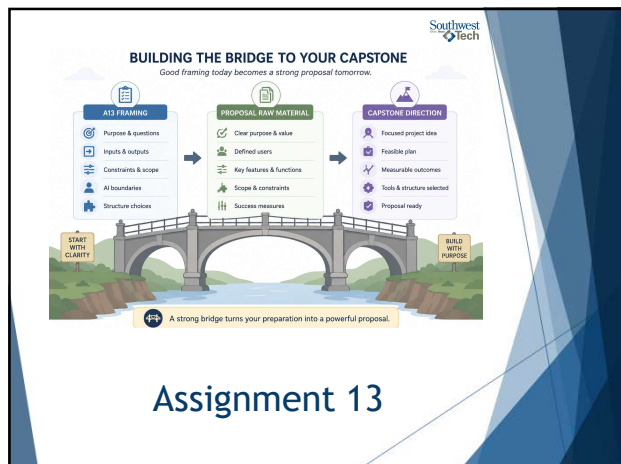
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Assignment #13

For Assignment 13, create a project framing document.

This can become the raw material for your capstone proposal, so make it concrete enough to review and revise.

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Evidence For A13

Your framing should include:

- ▶ project purpose
- ▶ inputs and outputs
- ▶ constraints
- ▶ likely structure
- ▶ risks or unknowns
- ▶ AI-use boundaries

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A13 EVIDENCE

Check each item you have addressed.

	PURPOSE I have a clear purpose and value for the project.	☐
	INPUTS AND OUTPUTS I have identified the key inputs and the expected outputs.	☐
	CONSTRAINTS I have listed the main limits or rules that shape the project.	☐
	LIKELY STRUCTURE I have a likely project structure that fits the purpose.	☐
	RISKS I have noted possible risks or challenges and early mitigation ideas.	☐
	AI BOUNDARIES I have defined how AI will be used as support, not replacement.	☐
	GOOD TO GO! If all boxes are checked, you are ready to move this framing into your capstone proposal.	☐

Evidence Example

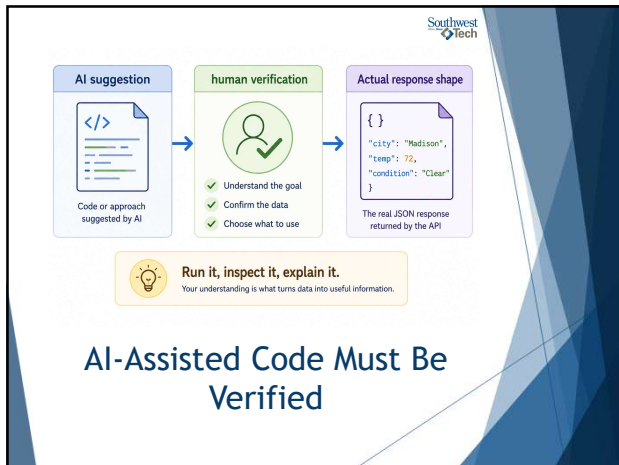
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AI Use

- ▶ AI can suggest code that looks believable but does not match the actual response.
- ▶ You must inspect the response, run the code, and explain what human decision you made.

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Closing Success Check

By the end of today, you should be able to say:

"I know what this project is for, what it needs, what it produces, what limits matter, and what AI should not decide for me."

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MY PROJECT OWNERSHIP

I own the plan. I guide the build.

WHAT IT IS FOR
Who is it for?
What problem or need does it address?

WHAT IT NEEDS
What are the key inputs, tools, or resources it requires?

WHAT IT PRODUCES
What outputs or results will it deliver?

WHAT AI SHOULD NOT DECIDE
What choices are mine as the owner?

I make the decisions.
AI helps with suggestions, checks, and support — not ownership.

Name: _____ Date: _____

Success Check

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Thank you!

Have Fun!

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